

COURSE COURSE

CODE:

NAME:

CSA14 Compiler

Design

L

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE OUTCOME COVERED

CO1: Apply lexical analysis for token specification and recognition using LEX tool. (BL3) Assessment Tool

Weightage: 40%

Duration

QUESTIONS: :60 Mins

5

On Class Total Marks:50

Annexure A

ANALYTICAL PROBLEM SOLVING

Code of Conduct:

I Mathusri P (192311363)_(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

[Problem

© Apply compiler
design phases for
27.

source program 10 = X
xy - 2 + 20

* lexical analyzer

204

<id 1> <= > <id ?> <*> <id 8>
<-> <id4> <+> <20>

*

Syntax Analyzes

§

* code optimization

idi

ide

ida

ssidy / +\

↓

tida 120.0

id = id2 * id 3 £ ti

generation

* Code 20LDE R2 idy

* Sematic **Analyze**

id,

Tid2

Dinida

*

idu

ADD Rang

LDE Ridz a MULE R1id3

SUBE Ruidy an

ADDE Ri #20.0

smeride Blo

int to flood

Intermediate Code generation

t1 = idy + int to float

(20): +2=id 3-ti

to todel

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[Problem

© Apply compiler
design phases for
27 .

source program 10 = X * Y

-2 : +20

* lexical analyzer

τ

<id 1> <=> <id 2> <*> <id> <->

<id4> <+> <20>

*

Syntax Analyzes

* code optimization

't1 = idy £20.0

idi

idq

+

↓

ida

*

sidy + x id4/

id = id2 * id 3 ti

i

* Code generation

idy

ADD R2

20 FLDE R2

* Sematic Analyzer

LDE R1idz

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(v)

#20.0

+Seride Ring of
idy

int to flood

千年
20

* Intermediate Code generation

t1 = idy + int to
to float (20)

(20):

+2 = id3 - £1

+3 = id2 t2

id1 = t3

3)

Ci

Ci

Page

Date

<+> <20>

sition Co

L

two

@

Construct Reg subsets

con

Regular expression

(Regex) for following

i) all strings beginning
and ending with's

(ii)

5, 11, 101, 111, 1001,

10113

1 (0+1)* 1+1*(nd no fuck Code"

01

(τ) All strings
containing the
cabshings of

201, 101, 010, 1101,

00110}

$(0+1)^*01(0+1)^*$

o'

*Cat all strings with
exactly one of
cit*

$\{0, 01, \underline{10}, 1110111\}$

Croqu

$1\underline{01}^*0$

do

(is) all strings with at least
 2 1 's

$\{ \underline{11}, 101, \underline{110}, 1001, 11011 \}$

$(0+1)^*1(0+\underline{1})^*\underline{1}(0+1)^+$

($\forall$) all strings **that** do
not **end** with 0

{ {, 1, 11, 101 11101 }
 $((0+1)^*1+8$

C construct and

Describe the

language

Generated by

The following Regular Expression.

(1) $(\epsilon + aa)^*$

1. $\epsilon, aa, agad,$
 $aaaaada \dots \}$ evento of
strings

shuhait

(i) $(a^* + b^*) * abb.$

$L - \{ abb, aabb, babb,$

aaabb, bbabb } and's
with abb

e

a cab) * abalipad bintang

1. { aba, aaba, baba, aaaba,
bbaba } ed ciber

(iv) (a tab)*

(V)

→

1. { ε, a, ab, aa, aab, abab, abaab?

Caba)*+ ca*b*) *

caba) * - o can more repetitions" of "abe" Ex: { ϵ , aba, abaaba, aboaberaba, }

(ab) - 0 (on) more repetition of any number of b ' s

Ex . { &, a, b, ab, aga bb, cabb, ciaabbb, abab, 3

+ ' means 'OR Cunion)

1-{\epsilon, aba, abaaba, a; b, ab, ad, bb, aabb, Gaabbb, abbaub. 3

pitogram

Q

Lex

to identify and
keywords!

algorithms

$\circ \rightarrow \tau$

101

program

Qstart the lex

2

3

users

@ Read the input
from the user onids
check whether the
input is a keyword
@ if it is a byword,
déplay "beyard"
3 otherwise check whether
it is a valid identifier
HiC if valid display
"identifier" on histones

@if @ otherwise

display "invalid

identifiers":"

6 Mop

Lox

program.

• 0 2

0+3°

ij

```
#include <stdio.h>
```

```
%%3
```

```
dd
```

```
Date
```

```
%%
```

```
int float char double
```

```
lifebel while for
```

```
return _ {Printf  
("keyword \n") _; } ughte  
[ _a-zA-Z ] [ _a-zA-Z0-9-_  
*  
_
```

```
%%  
{_printf ("Identifier \n");}  
9-04
```

doo

n}

```
{ printf ("Invalid Identifien");}
```

```
int main()
```

```
{
```

```
Op:
```

```
yydex (23
```

```
i/p: int
```

```
return 0; 4 :
```

```
D;
```

```
mirde op keyword
```

```
Vr3
```

```
3
```

```
wint yywrap!)
```

```
ξ
```

9-93

y

(1)

tifies

return 15

C **Prove** the
following

algebras
properties of
regular
expression.

$$\underline{E} + R = R$$

E

do dos

by the
idempotent law the

closure of a
language with
itself is the same
Language

$R^+ = R$ Example:

Let $R = ab$.

home

$R^+ = R^*$

$R^+ = \{ab\} \cup \{ab\} \cup \dots$

$\{ab\}$

Ach

0-90

11

(i) $R + \underline{D} = \underline{R}$

Stings

By the identity law

The ecoply set intours

no

Example.

Let $R = \{a, ab\}$

A

$\{a, ab\} \cup = \{a \text{ } \underline{aby} \text{ } \text{bilwal"}\}$

$\#hair"$

henie

$\underline{R+D} = R/$

(V)

- Cilli

i)

$$R2 = \varepsilon R = R$$

2 is the identity element
for

proof: The empty string &

is the

concatenation

and

$ER = R$

$R\varepsilon = R$

Example

$R = abra$

duan let $R = ab$.

Let $R = ab$ Mortarlo

privalo

$ab\varepsilon$ = ab

$Eab = ab$

bence

$R\varepsilon = ER = R/11$

$(ev) R\$ = \underline{OR} = 0$

proof

9-818

concatenating

any language

with the engi

Strings

set produces

produces *no*

RD = 0 ØR = 0 Example

ò

10

{**ab**} . \$ hence

R1 = \$R = 0/1

(1) R+S = S+Rhod

proof

anion is commutative, so

chang The Jarquage

does not

$$R+S = S+R$$

Example

$$R = \{a\}, \quad S = \{b\}$$

Let

for

then i

$Rts = \{ab\}$

$St12 = \&b, ay = da.b\}$:

Lonce

$Rts \Rightarrow +R?$

lovi

colo ban

Page

Date

changing the order

ai

care? of red
bomos ofe

RUBRICS FOR CALCULATION:

Criteria

Weightage

Level 4 Level 3 Exemplary Proficient

Level 2

-Level 1

Developing

Beginning

Problem

20%

Clearly

Understands Partial

Misinterprets

Understanding

interprets

problem,

minor gaps

problem with understanding; or unable to

misses key understand

identifies

elements

problem

all

parameter

s

and

requireme

nts

accurately

Method Selection

20%

Selects

Selects

most

correct

Inappropriate or Unable to

partially correct select

appropriat

method with

method

method

e

minor errors

method/for

mula with

strong

justificatio

n

Solution Process

25%

Logical,

Mostly

Disorganized

No

clear

systematic correct steps steps with
steps; multiple process

correct

with minor mistakes

errors

approach

throughout

Accuracy Calculation

of 20%

All

calculation

s

Minor calculation

errors

Frequent errors; No

valid

incorrect final calculations

answer

accurate;

correct final

Interpretation of 15% Results

answer

Clearly interprets results

Basic

interpretatio

Limited unclear

or No

interpretation

in with minor interpretation

with units

omissions

and

engineerin

g

relevance